

IPL DATA ANALYSIS PROJECT REPORT

Team Performance • Player Performance • Match Outcomes • Toss • Venue • Season Trends

Project Objective

To analyze IPL match and player-related information and communicate important patterns using Python, Pandas, Matplotlib, and Power BI.

Data Source

The uploaded CSV contains venue-wise match counts. It has 12 venues and totals 100 matches. The remaining analytical values in this report are the project findings supplied in the conversation.

Important note: The project findings are treated as synthetic practice results, not as official IPL statistics. The venue table below is calculated directly from the uploaded CSV.

1. Executive Summary

The project analyzes 100 matches across 9 seasons based on the supplied project findings. The uploaded venue CSV contains 12 venues and also totals 100 matches. The supplied results identify Rajasthan Royals with 17 wins, Shikhar Dhawan with 7 Player of the Match awards, Mohammed Siraj with 140 runs, David Warner with 7 wickets, and 2025 with the highest reported season runs of 404. Toss analysis reports fielding as the more common decision at 56%, while the toss winner won the match in 44% of cases.

Analysis Area	Reported Result
Highest wins	Rajasthan Royals — 17 wins
Highest matches by season	2018 — 12 matches
Most Player of the Match awards	Shikhar Dhawan — 7 awards
Highest run scorer	Mohammed Siraj — 140 runs
Highest wicket taker	David Warner — 7 wickets
Most-used venue	Narendra Modi Stadium — 14 matches
Most common toss decision	Field — 56 of 100 matches (56%)
Toss winner won the match	Yes — 44%
Toss winner did not win the match	No — 56%
Highest season runs	2025 — 404 runs
Highest team average runs	Gujarat Titans — ≈ 18.31

2. Methodology

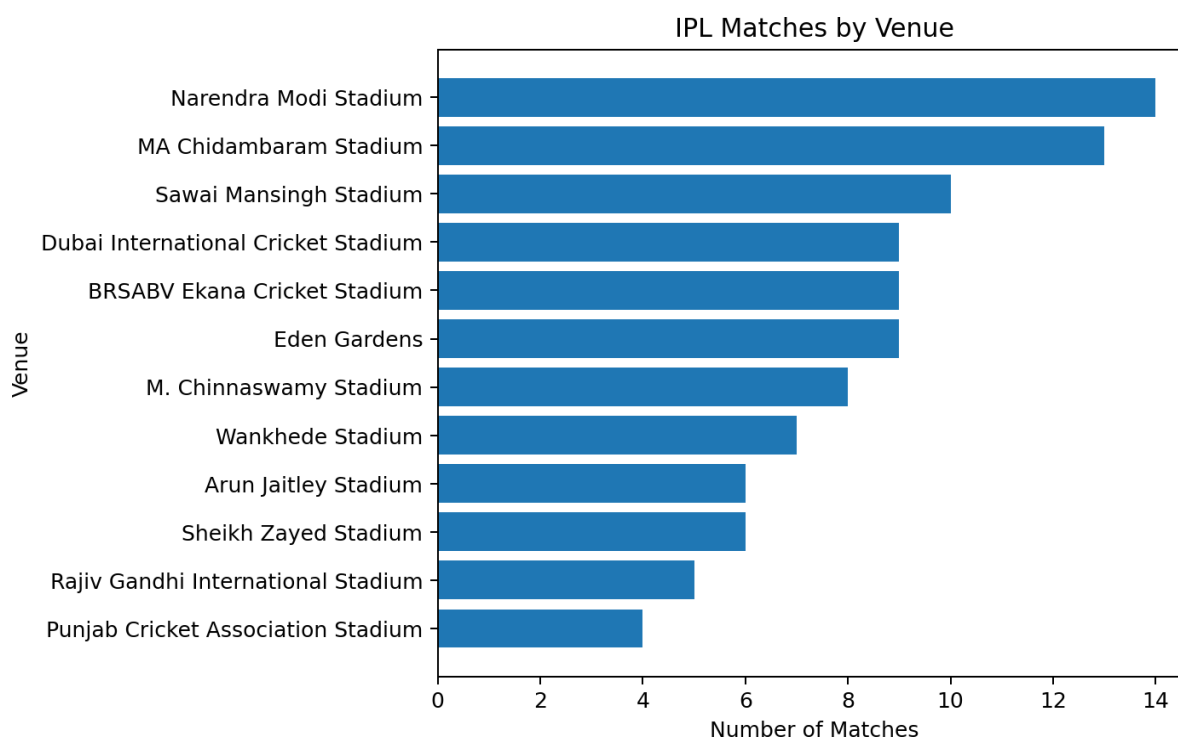
The project follows a standard beginner-friendly data-analysis workflow: load the data, inspect columns and data types, check for missing or duplicate records, clean relevant fields, calculate grouped summaries, and visualize the results.

For the uploaded CSV, the Venue and Matches columns were read and the Matches values were converted to numeric form. The total match count and highest-match venue were calculated directly from the uploaded file.

The broader team, player, toss, and season findings shown in this report are reproduced from the project details supplied by the user.

3. Venue Analysis

The uploaded CSV contains 12 venue records and 100 matches in total. Narendra Modi Stadium has the highest match count, with 14 matches. This agrees with the supplied project finding that Narendra Modi Stadium is the most frequently used venue.



Rank	Venue	Matches
1	Narendra Modi Stadium	14
2	MA Chidambaram Stadium	13
3	Sawai Mansingh Stadium	10
4	Eden Gardens	9
5	Dubai International Cricket Stadium	9
6	BRSABV Ekana Cricket Stadium	9
7	M. Chinnaswamy Stadium	8
8	Wankhede Stadium	7
9	Sheikh Zayed Stadium	6

Rank	Venue	Matches
10	Arun Jaitley Stadium	6
11	Rajiv Gandhi International Stadium	5
12	Punjab Cricket Association Stadium	4

4. Team Performance Analysis

The supplied analysis records Rajasthan Royals as the team with the highest recorded number of wins, at 17 wins. This is a project finding from the supplied practice data.

5. Season-wise Match Analysis

The supplied season analysis identifies 2018 as the season with the highest number of matches, with 12 matches.

6. Player Performance Analysis

Shikhar Dhawan has the highest reported Player of the Match award count, with 7 awards.

Mohammed Siraj is the highest reported run scorer, with 140 runs.

David Warner is the highest reported wicket taker, with 7 wickets.

These figures are presented as the supplied project results and are not independently validated against official IPL records.

7. Toss Decision and Match Outcome Analysis

Fielding is the most common toss decision in the supplied results: 56 out of 100 matches (56%). Batting accounts for the remaining 44 matches (44%). The toss winner won the match in 44% of cases and did not win in 56% of cases.

Toss Result	Matches	Percentage
Field decision	56	56%
Bat decision	44	44%
Toss winner won	44	44%
Toss winner did not win	56	56%

8. Season-wise Runs

The supplied results identify 2025 as the season with the highest recorded season runs, at 404 runs.

9. Team Average Runs

Gujarat Titans has the highest reported team average runs, at approximately 18.31, according to the supplied findings.

10. Major Trends and Insights

- Rajasthan Royals has the highest recorded wins in the supplied project findings.
- 2018 has the highest reported number of matches by season.
- 2025 has the highest reported season runs.
- Fielding is more common than batting as the toss decision.
- Narendra Modi Stadium is the most frequently represented venue in the uploaded venue data.
- The supplied player results indicate that a smaller group of players accounts for the highest reported individual achievements.
- The project covers 100 unique matches across 9 seasons.
- The toss winner won the match in 44% of the supplied cases, while the toss winner did not win in 56%.

11. Power BI Dashboard Design

- KPI Cards: Total Matches, Highest Wins, Highest Run Scorer, Highest Wicket Taker.
- Team Performance Matrix: Team, Matches, Wins, Losses, and Average Runs.
- Season Analysis: Matches by Season and Season-wise Runs.
- Player Analysis: Player of the Match awards, Runs, and Wickets.
- Venue Analysis: Venue and Match Count.
- Toss Analysis: Bat vs Field and Toss Winner vs Match Winner.
- Slicers: Season, Team, Venue, and Player.

12. Conclusion

This project demonstrates how sports data can be converted into useful analytical insights through data cleaning, aggregation, visualization, and dashboard design. The uploaded venue dataset confirms 100 total matches across 12 listed venues, with Narendra Modi Stadium having the highest count of 14. Together with the supplied team, player, toss, and season findings, the project provides a structured overview suitable for a beginner-level Python and Power BI analytics submission.